

DD_PLM-CFG-07 - Voice Tariff Catalog

Version: 1.0 Status: Draft for review Bundle: 09_catalogs Audience: catalog developers, billing developers, usage rating developers, QA, product operations Related DDs:

- DD_PLM-CFG-01 - Service Catalog
- DD_SIP-01 - Package Management
- DD_PLM-CFG-02 - Tax Configuration
- DD_BIL-05 - Wallet and Topup
- DD_MED-01 - Usage Mediation
- DD_RAT-01 - Usage Rating
- DD_REP-01 - Reporting Data Mart and Dashboard API

1. Purpose

PLM-CFG-07 defines the configurable voice tariff catalog used to rate voice usage records.

It answers these questions:

- Which tariff plan applies to a voice service or package?
- Which destination zone does a called number belong to?
- What rate applies for that zone, time period, and call type?
- What rounding increment is used?
- Which calls are free, blocked, zero-rated, or premium-rated?
- Are there bundled free minutes or allowances?

This DD is required before usage mediation and rating can be implemented. MED-01 will normalize CDRs. RAT-01 will rate those mediated CDRs. PLM-CFG-07 supplies the tariff configuration that rating uses.

2. Ownership Boundary

Concern	Owner
Voice tariff plans, destination zones, prefix mapping, time bands, per-minute rates, increments, allowances	PLM-CFG-07
Customer/package/service composition	SIP-01 and PLM-CFG-01
Raw CDR ingestion from VoipSwitch or other voice platforms	MED-01
CDR normalization, deduplication, quarantine	MED-01
Applying tariff to mediated CDR and producing charge/debit events	RAT-01
Wallet debit for PREPAID usage	BIL-05
POSTPAID usage billable event/invoice charge	BIL-01 / BIL-02
Tax calculation	PLM-CFG-02
Invoice/tax invoice generation	BIL-02 / BIL-02-TAX-01

Rule: PLM-CFG-07 never rates a CDR by itself. It exposes catalog data and validation APIs. RAT-01 performs runtime rating.

3. When To Use This Catalog

Use PLM-CFG-07 when:

- A new voice product is launched.
- A voice package needs a different local/international rate plan.

- A destination prefix changes zone.
- A new international destination zone is added.
- Peak/off-peak pricing changes.
- Emergency, toll-free, or premium numbers need special treatment.
- Product wants to add monthly free minutes or zone-specific allowance.

Do not use this catalog for:

- Broadband recurring package prices. These stay in SIP-01.
- Equipment fees. These stay in PLM-CFG-06.
- One-shot fees. These stay in BIL-CFG-01.
- Discounts and campaigns. These stay in PLM-CFG-04, SIP-03, SIP-05, and DIS-OP-01.

4. High Level Rating Relationship

```

flowchart LR
    VS["VoipSwitch / SIP platform"]
    MED["MED-01<br/>Usage Mediation"]
    RAT["RAT-01<br/>Usage Rating"]
    TAR["PLM-CFG-07<br/>Voice Tariff Catalog"]
    TAX["PLM-CFG-02<br/>Tax Config"]
    BIL["BIL-01 or BIL-05"]

    VS --> MED
    MED --> RAT
    RAT --> TAR
    RAT --> TAX
    RAT --> BIL

```

5. Core Concepts

5.1 Tariff Plan

A tariff plan is the rated price structure for a voice service.

Example:

```
WIK_RES_VOICE_STANDARD_2026
```

It may contain rates for local mobile, local fixed, on-net, East Africa, international, premium, toll-free, and emergency destinations.

5.2 Destination Zone

A destination zone groups dialed numbers that share rating behavior.

Examples:

- WIK_ON_NET
- KE_MOBILE
- KE_FIXED
- EA_REGION
- INTERNATIONAL_ROW
- TOLL_FREE
- EMERGENCY
- PREMIUM

5.3 Prefix Mapping

Prefix mapping converts a normalized called number into a destination zone using longest-prefix match.

Example:

```
+2547 -> KE_MOBILE
+25420 -> KE_FIXED
999 -> EMERGENCY
```

5.4 Rate

A rate defines price per unit for a tariff plan, destination zone, call type, time band, and effective date.

Example:

```
KE_MOBILE, PEAK, 3.00 KES per minute, 60/60 billing increment
```

5.5 Allowance

An allowance is an included quantity, such as 100 on-net minutes per month.

Allowance consumption is performed by RAT-01, but the allowance definition is stored here.

6. Rating Rules

Rule ID	Rule	Notes
R-VOICE-TAR-01	Tariff data is operator-scoped.	A prefix/rate for WIK does not apply to another operator.
R-VOICE-TAR-02	Tariff plans are versioned with effective dates.	Historical CDRs must rate with the tariff active at call time.
R-VOICE-TAR-03	Prefix resolution uses longest-prefix match.	+254711 beats +2547 if both exist.
R-VOICE-TAR-04	Emergency calls are zero-rated and must not consume allowance.	Regulatory safety.
R-VOICE-TAR-05	Toll-free calls are zero-rated to customer unless operator config marks them chargeable.	Default zero customer charge.
R-VOICE-TAR-06	Premium numbers require explicit zone and rate configuration.	Avoid accidental normal-rate billing.
R-VOICE-TAR-07	If no prefix/zone/rate is found, rating must quarantine the CDR.	Do not silently rate at zero.
R-VOICE-TAR-08	Rates cannot overlap for same plan, zone, call type, time band, and effective period.	Prevent ambiguous rating.
R-VOICE-TAR-09	Rounding increment is part of the rate row.	Supports per-destination billing increments.
R-VOICE-TAR-10	Allowance eligibility is explicit by tariff plan and destination zone.	Do not assume all minutes consume allowance.
R-VOICE-TAR-11	Currency is explicit and must match operator/package currency.	Kenya first uses KES.
R-VOICE-TAR-12	Tariff changes require future effective dates unless an elevated admin approves backdating.	Protect billing audit.

7. Data Model

7.1 voice_tariff_plan

Stores a commercial voice tariff plan.

Use this table when creating a new voice pricing structure or versioning an existing plan.

Column	Type	Purpose	Sample
tariff_plan_id	UUID PK	Internal plan id.	vtp-001
operator_code	TEXT	Operator/tenant.	WIK
tariff_plan_code	TEXT	Stable plan code used by packages/rating.	WIK_RES_VOICE_STD_2026
display_name	TEXT	Backoffice display name.	Residential Voice Standard 2026
billing_mode	TEXT	PREPAID, POSTPAID, BOTH.	BOTH

Column	Type	Purpose	Sample
currency_code	TEXT	Currency.	KES
status	TEXT	DRAFT, ACTIVE, RETIRED.	ACTIVE
effective_from	TIMESTAMP	First time this plan can be used.	2026-06-01T00:00:00Z
effective_to	TIMESTAMP NULL	End time, null means open.	NULL
created_at	TIMESTAMP	Creation timestamp.	2026-05-31T09:00:00Z
updated_at	TIMESTAMP	Last update timestamp.	2026-05-31T09:00:00Z

Recommended indexes:

- Unique (operator_code, tariff_plan_code).
- (operator_code, status, effective_from).

7.2 voice_destination_zone

Stores rating zones for voice destinations.

Use this table when defining destination groups such as local mobile, international, emergency, or premium.

Column	Type	Purpose	Sample
zone_id	UUID PK	Internal zone id.	vdz-001
operator_code	TEXT	Operator/tenant.	WIK
zone_code	TEXT	Stable destination zone code.	KE_MOBILE
zone_name	TEXT	Display name.	Kenya Mobile
zone_type	TEXT	ON_NET, NATIONAL, REGIONAL, INTERNATIONAL, TOLL_FREE, EMERGENCY, PREMIUM.	NATIONAL
default_charge_policy	TEXT	CHARGEABLE, ZERO_RATED, BLOCKED, QUARANTINE.	CHARGEABLE
status	TEXT	ACTIVE, RETIRED.	ACTIVE
created_at	TIMESTAMP	Creation timestamp.	2026-05-31T09:05:00Z
updated_at	TIMESTAMP	Last update timestamp.	2026-05-31T09:05:00Z

Recommended indexes:

- Unique (operator_code, zone_code).
- (operator_code, zone_type, status).

7.3 voice_destination_prefix

Maps normalized dialed-number prefixes to destination zones.

Use this table for longest-prefix match during rating.

Column	Type	Purpose	Sample
prefix_id	UUID PK	Internal prefix id.	vdp-001
operator_code	TEXT	Operator/tenant.	WIK
prefix	TEXT	Normalized prefix.	+2547
zone_id	UUID FK	Destination zone.	vdz-001
match_priority	INTEGER	Optional priority if equal-length conflict exists.	100
status	TEXT	ACTIVE, RETIRED.	ACTIVE
effective_from	TIMESTAMP	Prefix valid from.	2026-06-01T00:00:00Z
effective_to	TIMESTAMP NULL	Prefix valid until.	NULL
notes	TEXT NULL	Operational notes.	Safaricom mobile range

Recommended indexes:

- (operator_code, status, effective_from).

- (operator_code, prefix).

7.4 voice_time_band

Stores time-of-day and day-of-week bands used by rates.

Use this table when pricing has peak/off-peak or weekend differences.

Column	Type	Purpose	Sample
time_band_id	UUID PK	Internal id.	vtb-001
operator_code	TEXT	Operator/tenant.	WIK
time_band_code	TEXT	Stable code.	ANYTIME
display_name	TEXT	Display name.	Any time
days_of_week	TEXT	Comma-separated or JSON day set.	MON, TUE, WED, THU, FRI, SAT, SUN
start_time_local	TIME	Local start time.	00:00:00
end_time_local	TIME	Local end time.	23:59:59
timezone	TEXT	Operator timezone for band evaluation.	Africa/Nairobi
status	TEXT	ACTIVE, RETIRED.	ACTIVE

Recommended indexes:

- Unique (operator_code, time_band_code).

7.5 voice_tariff_rate

Stores the actual price for a plan, zone, call type, and time band.

Use this table as the primary input for RAT - 01 charge calculation.

Column	Type	Purpose	Sample
rate_id	UUID PK	Internal rate id.	vtr-001
operator_code	TEXT	Operator/tenant.	WIK
tariff_plan_id	UUID FK	Tariff plan.	vtp-001
zone_id	UUID FK	Destination zone.	vdz-001
time_band_id	UUID FK	Time band.	vtb-001
call_direction	TEXT	OUTBOUND, INBOUND, FORWARDED.	OUTBOUND
unit_type	TEXT	SECOND, MINUTE, CALL.	MINUTE
unit_price	DECIMAL(18,6)	Price per unit.	3.000000
setup_fee_amount	DECIMAL(18,6)	One-time fee per call.	0.000000
initial_increment_seconds	INTEGER	First billing increment.	60
subsequent_increment_seconds	INTEGER	Later increments.	60
minimum_charge_amount	DECIMAL(18,6)	Minimum charge after rounding.	0.000000
charge_policy	TEXT	CHARGEABLE, ZERO_RATED, BLOCKED, QUARANTINE.	CHARGEABLE
taxable_kind	TEXT	Tax lookup kind for PLM-CFG-02.	SERVICE
taxable_ref	TEXT	Tax lookup reference.	svc_VOIP_STD
effective_from	TIMESTAMP	Rate valid from.	2026-06-01T00:00:00Z
effective_to	TIMESTAMP NULL	Rate valid until.	NULL
status	TEXT	DRAFT, ACTIVE, RETIRED.	ACTIVE

Recommended indexes:

- (tariff_plan_id, zone_id, time_band_id, effective_from).

- (operator_code, status, effective_from).

7.6 voice_tariff_allowance

Stores included free or bundled minutes.

Use this table when a plan includes monthly or cycle-based voice allowance.

Column	Type	Purpose	Sample
allowance_id	UUID PK	Internal allowance id.	vta-001
operator_code	TEXT	Operator/tenant.	WIK
tariff_plan_id	UUID FK	Tariff plan.	vtp-001
allowance_code	TEXT	Stable allowance code.	ONNET_100_MIN
allowance_name	TEXT	Display name.	100 on-net minutes
included_seconds	INTEGER	Included seconds per cycle.	6000
eligible_zone_codes_json	JSONB	Zones that consume this allowance.	["WIK_ON_NET"]
cycle_policy	TEXT	MONTHLY, BILL_CYCLE, TOPUP_VALIDITY.	BILL_CYCLE
carry_over_policy	TEXT	NONE, ONE_CYCLE, CONFIGURED.	NONE
priority	INTEGER	Consumption priority.	10
status	TEXT	ACTIVE, RETIRED.	ACTIVE
effective_from	TIMESTAMP	Allowance valid from.	2026-06-01T00:00:00Z
effective_to	TIMESTAMP NULL	Allowance valid until.	NULL

Recommended indexes:

- Unique (operator_code, tariff_plan_id, allowance_code).

7.7 voice_tariff_binding

Binds a tariff plan to a package, service, or explicit subscription override.

Use this table when deciding which tariff plan applies during rating.

Column	Type	Purpose	Sample
binding_id	UUID PK	Internal binding id.	vtbnd-001
operator_code	TEXT	Operator/tenant.	WIK
binding_scope	TEXT	PACKAGE, SERVICE, SUBSCRIPTION_OVERRIDE.	PACKAGE
binding_ref	TEXT	Package id/service id/subscription id.	pkg_INET_VOICE_STD
tariff_plan_id	UUID FK	Tariff plan.	vtp-001
priority	INTEGER	Higher priority wins if multiple apply.	100
status	TEXT	ACTIVE, RETIRED.	ACTIVE
effective_from	TIMESTAMP	Binding valid from.	2026-06-01T00:00:00Z
effective_to	TIMESTAMP NULL	Binding valid until.	NULL

Recommended indexes:

- (operator_code, binding_scope, binding_ref, status).
- (tariff_plan_id, status).

8. APIs

8.1 Tariff Plan APIs

```
GET /api/plm/voice-tariff-plans
POST /api/plm/voice-tariff-plans
GET /api/plm/voice-tariff-plans/{tariff_plan_id}
PATCH /api/plm/voice-tariff-plans/{tariff_plan_id}
POST /api/plm/voice-tariff-plans/{tariff_plan_id}/activate
POST /api/plm/voice-tariff-plans/{tariff_plan_id}/retire
```

8.2 Zone And Prefix APIs

```
GET /api/plm/voice-destination-zones
POST /api/plm/voice-destination-zones
```

```
GET /api/plm/voice-destination-prefixes
POST /api/plm/voice-destination-prefixes
POST /api/plm/voice-destination-prefixes/bulk-import
```

8.3 Rate APIs

```
GET /api/plm/voice-tariff-rates?tariffPlanCode={code}
POST /api/plm/voice-tariff-rates
POST /api/plm/voice-tariff-rates/bulk-import
POST /api/plm/voice-tariff-rates/validate-overlap
```

8.4 Rating Lookup API

Used by RAT-01.

```
POST /api/plm/voice-rating/lookup
```

Request:

```
{
  "operatorCode": "WIK",
  "subscriptionId": "SUB-700045",
  "packageRef": "pkg_INET_VOICE_STD",
  "serviceRef": "svc_VOIP_STD",
  "calledNumberNormalized": "+254711222333",
  "callDirection": "OUTBOUND",
  "callStartedAt": "2026-06-02T10:15:00+03:00"
}
```

Response:

```
{
  "tariffPlanCode": "WIK_RES_VOICE_STD_2026",
  "zoneCode": "KE_MOBILE",
  "timeBandCode": "ANYTIME",
  "chargePolicy": "CHARGEABLE",
  "unitType": "MINUTE",
  "unitPrice": 3.0,
  "currency": "KES",
  "initialIncrementSeconds": 60,
  "subsequentIncrementSeconds": 60,
  "setupFeeAmount": 0.0,
  "minimumChargeAmount": 0.0,
  "taxableKind": "SERVICE",
  "taxableRef": "svc_VOIP_STD"
}
```

9. Sample Rating Calculation

Input CDR after mediation:

```
{
  "operatorCode": "WIK",
  "subscriptionId": "SUB-700045",
  "serviceRef": "svc_VOIP_STD",
  "calledNumberNormalized": "+254711222333",
  "durationSeconds": 75,
  "callStartedAt": "2026-06-02T10:15:00+03:00"
}
```

Catalog lookup returns:

```
unit_price = 3.00 KES per minute
initial_increment_seconds = 60
subsequent_increment_seconds = 60
```

Rating calculation performed by RAT-01:

```
75 actual seconds -> rounded to 120 billable seconds
120 / 60 = 2 billed minutes
2 * 3.00 = 6.00 KES
```

PLM-CFG-07 supplies the data. RAT-01 performs this calculation and creates the usage charge/debit.

10. Events

Event	Topic	When emitted
VoiceTariffPlanCreated	sophix.plm.voice-tariff.plan.created	Plan created.
VoiceTariffPlanActivated	sophix.plm.voice-tariff.plan.activated	Plan activated.
VoiceTariffPlanRetired	sophix.plm.voice-tariff.plan.retired	Plan retired.
VoiceTariffRateChanged	sophix.plm.voice-tariff.rate.changed	Rate created/updated/retired.
VoiceDestinationPrefixChanged	sophix.plm.voice-destination-prefix.changed	Prefix table changes.
VoiceTariffBindingChanged	sophix.plm.voice-tariff.binding.changed	Binding changes.

Consumers:

- RAT-01 invalidates tariff cache.
- REP-01 tracks catalog history and price changes.
- Backoffice refreshes configuration views.

11. Frontend Usage

FE-APP-01 Backoffice uses PLM-CFG-07 for:

- Voice tariff plan list/detail.
- Destination zone management.
- Prefix import and validation.
- Rate import and overlap validation.
- Allowance setup.
- Tariff binding to package/service.
- Activation/retirement workflow.

Self-care and sales apps normally do not manage tariffs. They may display customer-friendly price summaries from package APIs, not raw tariff tables.

12. Infra Recommendation

PLM-CFG-07 is a catalog service feature, not a high-throughput runtime service by itself.

Recommended medium deployment:

Component	Recommendation
API pods	Use existing PLM/catalog API pods. No dedicated service required unless catalog load grows.
DB	Existing catalog PostgreSQL schema.
Cache	Redis or local cache for rating lookup. RAT-01 should cache tariff lookup results with event invalidation.
Workers	Optional import validation worker for large prefix/rate uploads.

Recommended partitions:

- No partition required for core catalog tables at launch.
- Keep audit/history tables if catalog framework already supports them.
- Prefix imports should be batch-tracked for rollback and validation.

OS recommendation:

- Same as catalog service baseline: Ubuntu Server 24.04 LTS for VPS, or hardened Debian/Ubuntu slim images in containers.

13. AWS Deployment Recommendation

AWS component	Usage
ECS/EKS	Existing catalog API containers.
RDS PostgreSQL	Catalog tables.
ElastiCache Redis	Optional tariff lookup cache.
S3	Optional storage for uploaded prefix/rate CSV import files.
CloudWatch	Import failures, overlap validation errors, activation events.

No internet-facing callback is required for this DD.

14. Testing Requirements

14.1 Unit Tests

- Longest-prefix match picks most specific prefix.
- Rate overlap validation rejects ambiguous rates.
- Emergency zone returns zero-rated policy.
- Premium zone requires explicit rate.
- Effective-date lookup selects historical rate.
- Binding priority selects subscription override before package binding.

14.2 Integration Tests

- Create draft tariff plan, add zones, add prefixes, add rates, activate plan.
- Bulk prefix import rejects duplicate active prefixes.
- Rating lookup returns correct rate for package/service/called number/time.
- Rate change event invalidates rating cache.

14.3 End To End Tests With RAT-01

- 75-second mobile call rates as 2 billed minutes.
- Emergency call produces zero charge.
- Unknown prefix quarantines CDR in RAT-01.
- Package-bound allowance is consumed before paid rate.

15. Definition Of Done

PLM-CFG-07 is complete when:

- Tariff plan, zone, prefix, time band, rate, allowance, and binding tables are implemented.
- Admin APIs can create, validate, activate, and retire tariff plans.
- Rating lookup API returns deterministic tariff data.
- Longest-prefix match works.
- Effective-date rating lookup works.
- Rate overlap validation works.
- Events are emitted for cache invalidation.
- Backoffice can manage the catalog.
- RAT-01 can use the lookup API without reading catalog tables directly.